10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 2.2 — Graphical Solutions

Unit 2 | Solved Exercise | New Book (PECTAA)

Exercise 2.2 (Solved)

1. Find the points of intersection of the following linear equations with coordinate axes graphically:

(i) $x + y = 8$

(ii) $x - y = 1$

(iii) $x - 2y = 1$

(iv) $x - 2y + 2 = 0$

(v) $5x - 5y = 1$

A straight line always cuts the coordinate axes at two points. Put y = 0 to get the x-intercept, and put x = 0 to get the y-intercept. Joining these two points gives the graph.

(i) $x + y = 8$

Put y = 0

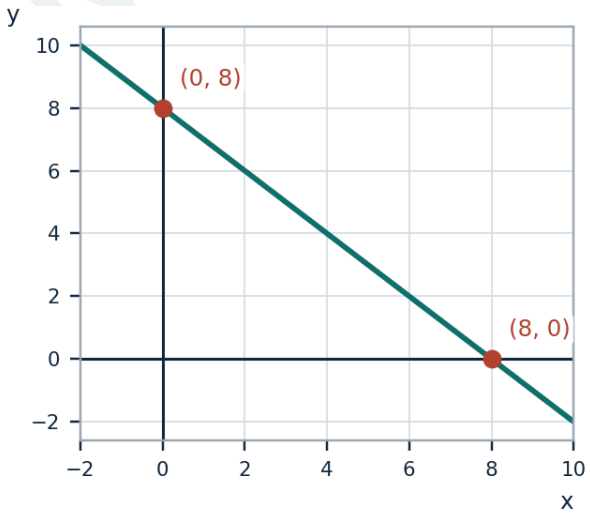
x = 8, so the line cuts the x-axis at (8, 0)

Put x = 0

y = 8, so the line cuts the y-axis at (0, 8)

table of values

x	8	0
y	0	8



Answer: x-intercept = (8, 0), y-intercept = (0, 8)

(ii) $x - y = 1$

Put y = 0

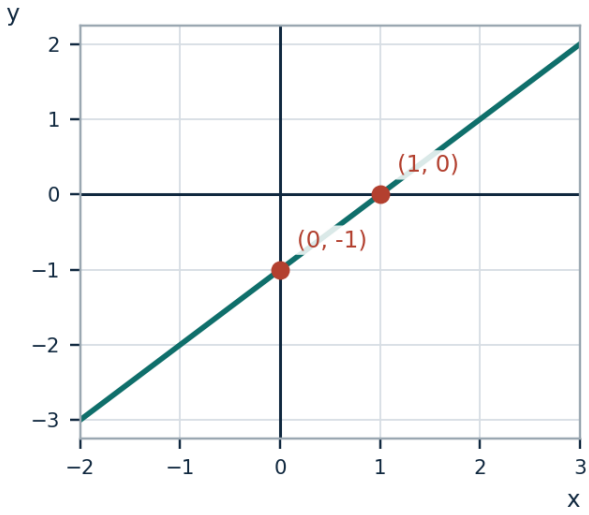
x = 1, so the line cuts the x-axis at (1, 0)

Put x = 0

y = -1, so the line cuts the y-axis at (0, -1)

table of values

x	1	0
y	0	−1



Answer: x-intercept = (1, 0), y-intercept = (0, -1)

(iii) x − 2y = 1

Put y = 0

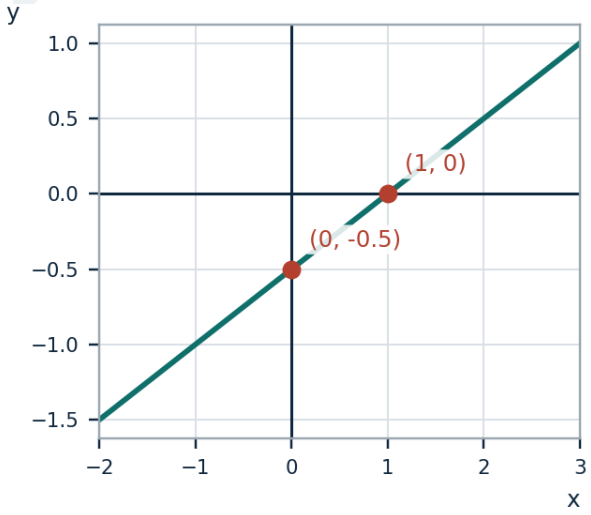
x = 1, so the line cuts the x-axis at (1, 0)

Put x = 0

y = -0.5, so the line cuts the y-axis at (0, -0.5)

table of values

x	1	0
y	0	−0.5



Answer: x-intercept = (1, 0), y-intercept = (0, -0.5)

(iv) x − 2y + 2 = 0

Put y = 0

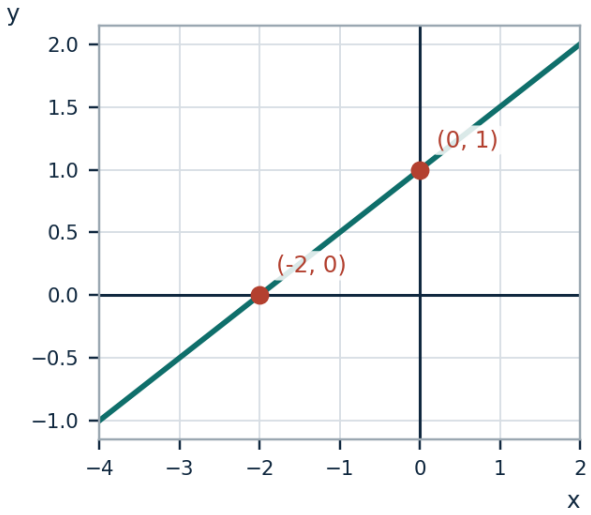
x = -2, so the line cuts the x-axis at (-2, 0)

Put x = 0

y = 1, so the line cuts the y-axis at (0, 1)

table of values

x	−2	0
y	0	1



Answer: x-intercept = (-2, 0), y-intercept = (0, 1)

(v) 5x − 5y = 1

Put y = 0

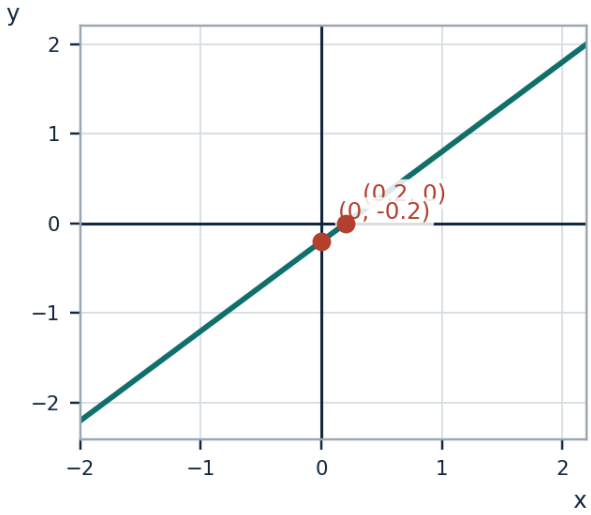
x = 0.2, so the line cuts the x-axis at (0.2, 0)

Put x = 0

y = -0.2, so the line cuts the y-axis at (0, -0.2)

table of values

x	0.2	0
y	0	−0.2



Answer: x-intercept = (0.2, 0), y-intercept = (0, -0.2)

2. Solve the following system of linear equations graphically:

(i) $x + y = 8,\ 3x - y = 4$

(ii) $x - y = 1,\ x + 2y = 7$

(iii) $x - 2y = 1,\ 2x + y = 2$

(iv) $y = 2x + 2,\ 3x + 2y = 4$

(v) $3y = 2x + 8,\ x + y = 1$

Method: draw both lines on the same graph paper. The point where the two lines intersect is the solution of the system (both equations are satisfied by that point).

(i) $x + y = 8$ and $3x - y = 4$

$x + y = 8$

$y = 8 - x$

$3x - y = 4$

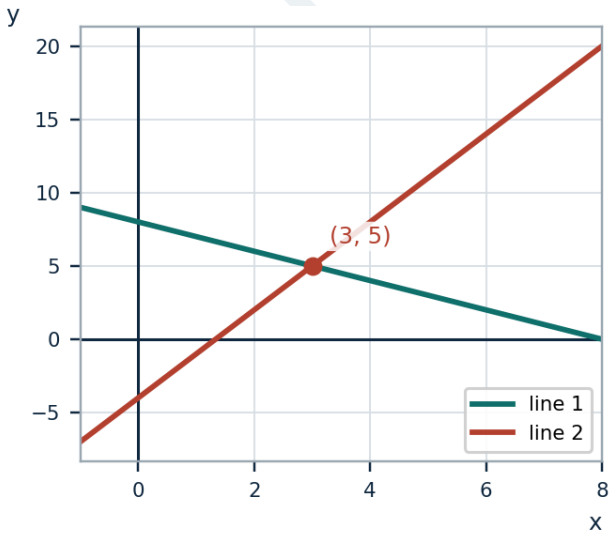
$y = 3x - 4$

y-values of line 1 ($x = 0, 3$)

x	0	3
y	8	5

y-values of line 2 ($x = 0, 3$)

x	0	3
y	−4	5



Both lines meet at (3, 5)

Answer: $x = 3, y = 5 \Longrightarrow$ solution set = {(3, 5)}

(ii) $x - y = 1$ and $x + 2y = 7$

$x - y = 1$

$y = x - 1$

$x + 2y = 7$

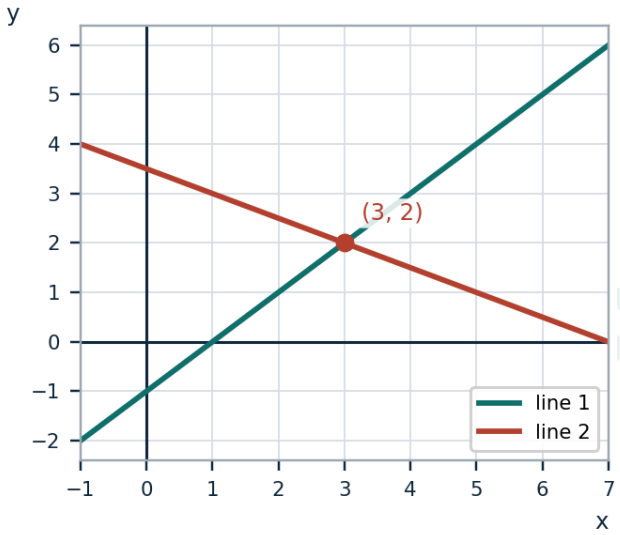
$y = (7 - x)/2$

y-values of line 1 (x = 0, 3)

x	0	3
y	−1	2

y-values of line 2 (x = 0, 3)

x	0	3
y	3.5	2



Both lines meet at (3, 2)

Answer: x = 3, y = 2 ⟹ solution set = {(3, 2)}

(iii) x − 2y = 1 and 2x + y = 2

x − 2y = 1

y = (x − 1)/2

2x + y = 2

y = 2 − 2x

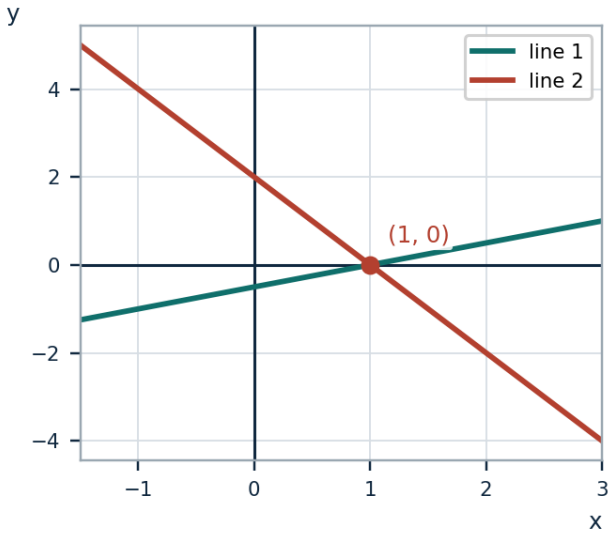
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y-values of line 1 (x = 0, 1)

x	0	1
y	−0.5	0

y-values of line 2 (x = 0, 1)

x	0	1
y	2	0



Both lines meet at (1, 0)

Answer: x = 1, y = 0 ⟹ solution set = {(1, 0)}

(iv) y = 2x + 2 and 3x + 2y = 4

y = 2x + 2

3x + 2y = 4

y = (4 − 3x)/2

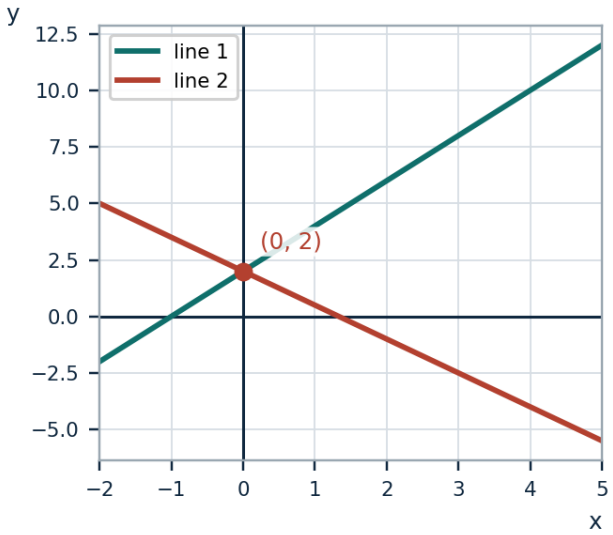
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y-values of line 1 (x = 0, 2)

x	0	2
y	2	6

y-values of line 2 (x = 0, 2)

x	0	2
y	2	−1



Both lines meet at (0, 2)

Answer: x = 0, y = 2 ⟹ solution set = {(0, 2)}

(v) 3y = 2x + 8 and x + y = 1

3y = 2x + 8

y = (2x + 8)/3

x + y = 1

y = 1 − x

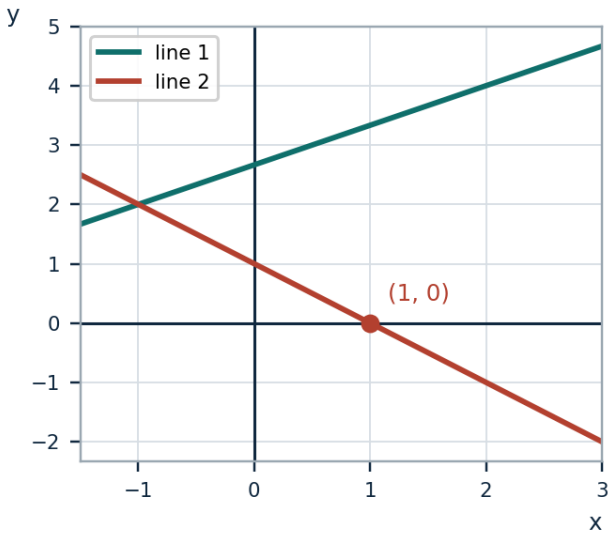
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y-values of line 1 (x = 0, 2)

x	0	2
y	2.6666666666666665	4

y-values of line 2 (x = 0, 2)

x	0	2
y	1	−1



Both lines meet at (1, 0)

Answer: x = 1, y = 0 ⟹ solution set = {(1, 0)}

3. Solve the following equations graphically:

(i) $y = 8x - 32,\ y = x^2 - 6x + 8$

(ii) $y + x = 2,\ y = 2x^2 + x - 10$

One equation is a straight line and the other is a parabola. Draw both graphs; the x-coordinates of their points of intersection are the solutions.

(i) $y = 8x - 32$ and $y = x^2 - 6x + 8$

For the line $y = 8x - 32$:

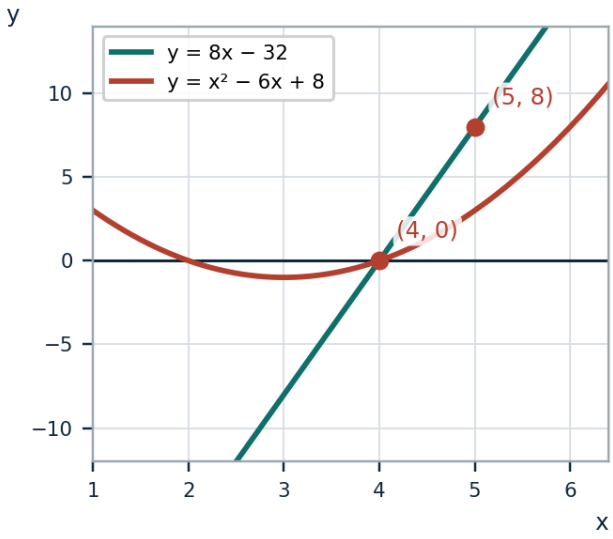
line

x	2	4	5
y	−16	0	8

For the curve $y = x^2 - 6x + 8$:

parabola

x	0	2	3	4	5	6
y	8	0	−1	0	3	8



The line meets the curve at (4, 0) and (5, 8)

Answer: x = 4, 5 ⟹ solution set = {4, 5}

(ii) $y + x = 2$ and $y = 2x^2 + x - 10$

Line: $y = 2 - x$

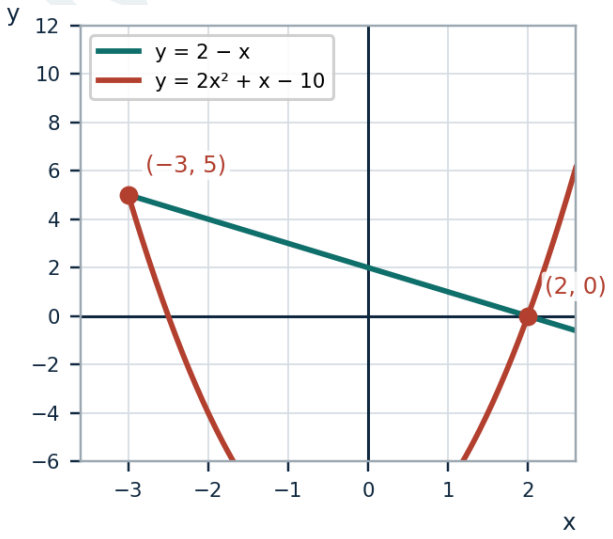
line

x	0	3	4
y	2	−1	−2

Curve: $y = 2x^2 + x - 10$

parabola

x	0	2	3	4
y	−10	0	11	26



The line meets the curve at (2, 0) and (−3, 5)

Answer: x = 2, −3 ⟹ solution set = {2, −3}